

BALANCE–FIELD FRAMEWORK (BFG)

The BFG Strong Universal Reclosure Equation

A Self-Closed Mathematical Architecture from Level-0 Formation to Domain-Independent Closure Classes and Natural-System Realization

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Scientific

status

Independent theoretical whitepaper and falsifiable research program. Exact operator identities are proved under their stated assumptions. The Strong Universal Reclosure construction is mathematically universal over the declared totalized admissible BFG state category: one fixed operator family generates domain-independent invariant classes, with failure mapped to a terminal non-generative object. This universality is conditional on the stated structural axioms and model-completion choices. Natural-system identifications with physical, chemical, biological, neural, conscious, or phenomenal regimes remain empirical realization hypotheses. The numerical studies are deterministic internal demonstrations, not empirical validation.

Abstract

This whitepaper develops the Strong Universal Form of the Balance-Field Framework (BFG). The central mathematical objective is stronger than reuse of one recursion across preselected domains: the framework is closed as a single admissible state category on which one level-independent reclosure operator acts as a self-map. Starting from Level-0 admissible formation, the construction uses the positive master load, the exact neutral resolvent pair $C_N(Y)=(I+Y)^{-1}$ and $B_N(Y)=Y(I+Y)^{-1}$, recursive persistence, order-indexed mediation, canonical retention/complement/export decomposition, and a universal formation gate [1–6]. The Strong Universal Form removes target-aware stage predicates from the generating mathematics. Instead, every iterate is characterized by a domain-independent invariant signature and belongs to an equivalence class in a universal quotient space. Physical, atomic, molecular, prebiotic, cellular, organismic, cognitive, conscious, and phenomenal labels are introduced only afterward as empirical realization maps into those already-generated classes. To close the principal formal gaps, this paper defines: (i) a totalized admissible state category with a terminal non-generative object; (ii) functorial transport of the trinitarian capacity operators into every next candidate packet; (iii) automatic preservation of positive master geometry through Gram-type construction; (iv) a prospectively fixed Riesz persistence projector with a spectral-separation requirement; and (v) a target-independent invariant quotient that prevents desired natural classes from entering the generator. Under these declared structural assumptions, the Universal Reclosure operator is an endomorphism of the admissible BFG category and its iterates require no sector-specific fundamental response law. Six deterministic numerical demonstrations test exact neutral partitioning, spectral generativity, order-feedback stability, shared-neutral integration, preserved differentiation, and bounded recursive trajectories. The mathematical universality claim is therefore precise: BFG is universal over its declared admissible state category. The separate statement that all natural systems realize that category remains a falsifiable natural-realization hypothesis rather than a theorem.

Keywords: Balance-Field Framework; BFG; strong universality; universal reclosure; Level-0; trinitarian formation; neutral resolvent; positive master geometry; recursive persistence; Riesz projector; endomorphism; invariant quotient; no-retuning; natural-system realization; abiogenesis; organismic closure; consciousness; qualia; falsifiability.

Contribution and source-line provenance

The Strong Universal Form, the totalized admissible state category, functorial capacity transport, target-independent invariant quotient, natural-realization commutation test, universality theorem, and numerical synthesis are the responsibility of Marcel Theodor Wendé. Earlier source lines are credited where their mathematics is used, while the present paper remains an independent unification construction.

- Marcel Theodor Wendé and Thomas W. Loker: co-authored Level-0 foundation source line used for pre-polar admissibility, determinate formation, and the Level-0 boundary [3].
 - Marcel Theodor Wendé and Luke Kenneth Casson Leighton: recursive conjugate-closure source line, including the form $R_C=MRM^\dagger$ with $M=\sqrt{N}$ used in the persistence architecture [4].

Reference and numbering convention

This whitepaper uses a single numerical reference sequence in order of first substantive appearance, with bracketed in-text citations and one numbered bibliography entry per source. This follows the professional sequential-numbering discipline used by major technical journals; the external bibliography was checked against publisher or authoritative indexing metadata where available on 8 September 2026. Equations, figures, and tables are each numbered sequentially throughout the manuscript.

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1. Purpose, scope, and the unification criterion

The central question is whether one mathematically closed BFG architecture can generate every admissible higher closure without inserting a new fundamental law, a target-aware classifier, or a sector-specific repair. The required notion of universality is therefore stronger than cross-domain resemblance. It requires one domain-independent state class, one fixed operator grammar, one positive-load construction, one persistence selection rule, one order-consistency rule, and one reclosure map that sends admissible objects back into the same class [1–6].

The Strong Universal Form distinguishes two claims. Mathematical universality means that the same reclosure operator is a self-map of the declared admissible BFG category and that no physical, biological, cognitive, or phenomenal label appears in its definition. Natural universality is the empirical claim that real systems can be mapped into the resulting invariant quotient without domain-specific retuning. The first claim is established conditionally from the declared structural assumptions; the second is deliberately left open to falsification.

$$\mathcal{T}_0 \rightarrow \Delta N_0 \rightarrow (P_1, N, P_2) \rightarrow K_0 \rightarrow \delta^* \rightarrow H_C \rightarrow Y_C \rightarrow \{C_N, B_N\} \rightarrow R_C \rightarrow \mathcal{U} \rightarrow \mathcal{U}^2 \rightarrow \dots \quad (1)$$

Equation (1) remains the red thread, but the meaning of $\mathcal{U}$ is strengthened. It is no longer merely a compact notation for repeated use of similar ingredients. It is completed below as a totalized, level-independent self-map whose output carries the same admissible structural data needed for the next iteration. Named natural regimes are downstream realization hypotheses, never terms in the generator.

Unification

criterion

An alleged stage transition is not counted as unified if it requires a new sector-specific fundamental response function, a post-hoc role redefinition, a target-aware threshold, or a carrier map chosen only after the higher-level outcome is known.

2. Epistemic discipline and claim-state hierarchy

The claim-state hierarchy is retained rigorously [1,2]. Definitions specify the state category and equivalence relation; inherited structural axioms supply Level-0, exact neutral resolution, positive closure geometry, and recursive persistence; the Strong Universal completion specifies how these structures are transported under reclosure; the mathematical theorem is conditional on those declared axioms; physical principles such as local physical closure remain separate; and all identifications with natural systems remain empirical hypotheses.

Table 1. Claim-state discipline used throughout the paper.

Class	Meaning
Definition	Introduces an object or equivalence class; true only by stipulation within the formalism.
Axiom / inherited BFG structure	A structural assumption imported from the canonical corpus.
Theorem	Exact mathematical consequence under explicitly stated assumptions.
Conditional theorem	Exact consequence after additional assumptions not yet derived upstream.
Physical principle	A universal physical bridge constrained by no-retuning; LPCP is the canonical example [1].
Unification construction	New in this paper: a fixed operator recursion built only from inherited canonical objects.
Model completion	A concrete representational or toy choice used to make a calculation executable.
Spectral prescription	A fixed readout from the master spectrum; numerical content must be computed.

Empirical hypothesis	Identification of a formal closure class with an observed physical or phenomenal regime.
Synthetic simulation	Numerically exact for the stated toy model; not empirical validation.

The strongest formal conclusion is therefore not that every natural object has already been derived. It is that, once the Strong Universal admissible category is fixed, the same target-independent map closes on that category and generates only domain-independent invariant classes. The separate burden is to show that atoms, cells, organisms, conscious states, or other natural systems realize those classes under prospectively fixed measurement maps.

3. Canonical source architecture

3.1 Level-0 trinitarian source

The canonical Level-0 formulation defines a finite-dimensional real Hilbert space V_0 with three primitive admissibility roles: difference capacity $\mathcal{D}_0$, coherence capacity $\mathcal{C}_0$, and neutral admissibility $\mathcal{N}_0$ [1,3].

$$\mathcal{T}_0 = (\mathcal{D}_0, \mathcal{C}_0, \mathcal{N}_0) \quad (2)$$

$$\mathcal{K}_0 = \mathcal{C}_0 + \mathcal{N}_0 - \mathcal{D}_0 \quad (3)$$

$$\mathcal{A}_{\text{Adm}}(\delta) = \frac{1}{2}\langle \delta, \mathcal{K}_0 \delta \rangle + \frac{1}{4}\|\delta\|^4, \quad \delta \in V_0 \quad (4)$$

A nontrivial isolated formation branch requires a simple negative minimum eigenvalue of $\mathcal{K}_0$ and a positive spectral gap.

$$\lambda_0 = \min \sigma(\mathcal{K}_0) < 0, \quad \text{mult}(\lambda_0)=1 \quad (5)$$

$$\Delta_0 = \lambda_1 - \lambda_0 > 0 \quad (6)$$

$$\delta^*_{\pm} = \pm \sqrt{-\lambda_0} e_0 \quad (7)$$

The quartic term makes the admissibility functional coercive; the Rayleigh–Ritz argument isolates the lowest eigenvector. The two minima are polarity-related. This is a mathematical formation result inside the stated real finite-dimensional model, not an empirical cosmological timestamp [1–3].

3.2 Exact neutral elimination

Let A map a difference channel into a neutral mediation channel. The canonical closure formulation uses a quadratic neutral completion whose auxiliary neutral variable can be eliminated exactly [1].

$$\mathcal{A}_{\text{C}}(D, \nu) = \frac{1}{2}\|D - A^\dagger \nu\|^2 + \frac{1}{2}\|\nu\|^2 \quad (8)$$

$$Y = A^\dagger A \geq 0 \quad (9)$$

$$\nu^* = (I + AA^\dagger)^{-1} A D \quad (10)$$

$$\mathcal{C}_{\text{N}}(Y) = (I + Y)^{-1} \quad (11)$$

$$\mathcal{B}_{\text{N}}(Y) = Y(I + Y)^{-1} = I - \mathcal{C}_{\text{N}}(Y) \quad (12)$$

$$\mathcal{C}_{\text{N}}(Y) + \mathcal{B}_{\text{N}}(Y) = I \quad (13)$$

$$0 < \mathcal{C}_{\text{N}}(Y) \leq I, \quad 0 \leq \mathcal{B}_{\text{N}}(Y) < I \quad (14)$$

$$\mathcal{A}_{\text{C}}^{\min}(D) = \frac{1}{2}\langle D, \mathcal{C}_{\text{N}}(Y) D \rangle \quad (15)$$

$$\frac{1}{2}\|D\|^2 = \frac{1}{2}\langle D, \mathcal{C}_{\text{N}} D \rangle + \frac{1}{2}\langle D, \mathcal{B}_{\text{N}} D \rangle \quad (16)$$

Equations (9)–(16) are the central exact identities of the unification. They supply a bounded retained response, a bounded complementary response, and an exact quadratic partition. Standard positive-matrix analysis supports the spectral and perturbative facts used here [7,8].

3.3 Positive master closure geometry and spectral sectors

The canonical universal closure architecture lifts the local neutral pair into a positive master closure geometry [1].

$$B_C = D_{cov} c \quad (17)$$

$$H_C = B_C^\dagger W_N B_C \geq 0 \quad (18)$$

$$Y_C = L_C^\dagger H_C L_C \geq 0 \quad (19)$$

Sector identity is licensed only by upstream spectral gaps. If S_C is the relevant self-adjoint master sector operator and Δ_a are disjoint spectral windows, then:

$$P_a = \mathbb{1}_{\{\Delta_a\}}(S_C), \quad P_a P_b = \delta_{ab} P_a, \quad \Sigma_a P_a = I \quad (20)$$

$$\beta_a = B_N^{\wedge\{1/2\}} P_a B_N^{\wedge\{1/2\}}, \quad \beta_a \geq 0, \quad \Sigma_a \beta_a = B_N \quad (21)$$

This positive partition is the exact no-double-counting backbone of later readouts. A claimed higher-stage channel that has no upstream gap or invariant distinction is not permitted to become a new fundamental BFG sector simply by naming it.

3.4 Recursive persistence and source-line continuity

The recursive conjugate-closure source line is preserved explicitly. Its conjugate form is used in the persistence architecture [4]. The canonical universal architecture uses a protected persistence direction plus a stable complement [1].

$$R_C = M U_\Omega C_E M^\dagger, \quad M = \sqrt{N_R} \quad (22)$$

$$R_C e_T = e_T \quad (23)$$

$$\|R_C|_{\{E_s\}}\|_L \leq r < 1 \quad (24)$$

$$P_T = \text{RieszProj}_{\{\lambda=1\}}(Dc_{\text{phys}}) \quad (25)$$

For $s = a e_T + s_s$, the stable component contracts at least geometrically, so the full trajectory remains power bounded. This motivates the carrier-neutral witness notion used below. The mathematical distinction between eigenvalue information and robust subspace information follows standard perturbation theory [7,8].

4. Order-indexed neutral mediation

A prior BFG formulation separates the inherited source ratio $\Phi_{\text{src},n}$ from the realized local order state φ_n , the neutral relation, the operator realization $\hat{N}_n[\varphi_n]$, and an observable scalar response ν_n [5]. This separation is retained because it prevents a single symbol from carrying incompatible roles.

$$\Phi_{\text{src},n} = (E_{\text{exp},n} I_n) / (E_{\text{bind},n} I_{\{0,n\}}) \quad (26)$$

$$\varphi_n \nu_n(\varphi_n) = \Phi_{\text{src},n} \quad (27)$$

$$\varphi_n = F_n(\varphi_n) := \Phi_{\text{src},n} / \nu_n(\varphi_n) \quad (28)$$

If F_n maps a closed complete positive domain into itself and has Lipschitz constant $L_n < 1$, the Banach contraction principle gives a unique local fixed point and convergent Picard iteration [5,9]. Cross-level coupling with induced Lipschitz gain κ remains jointly contractive under the sufficient condition:

$$L + \kappa < 1, \quad L = \max_n L_n \quad (29)$$

The order-indexed extension also distinguishes structural level n from physical/developmental time t . This distinction is essential: the formal hierarchy is an order of closure precedence, not a claim that all physical systems literally pass through the same temporal sequence.

5. Strong Universal Reclosure Architecture

5.1 Motivation

The previous reclosure construction removed explicit level-specific generativity maps but still left two possible sources of hidden freedom: how the next formation operator is obtained and how a desired natural stage is recognized. The Strong Universal Form closes both locations. The next trinitarian capacities are transported functorially from the current admissible object into the candidate packet, and all mathematical classification is performed in a target-independent invariant quotient. No label such as atom, cell, organism, consciousness, or quale enters the definition of $\mathfrak{U}$.

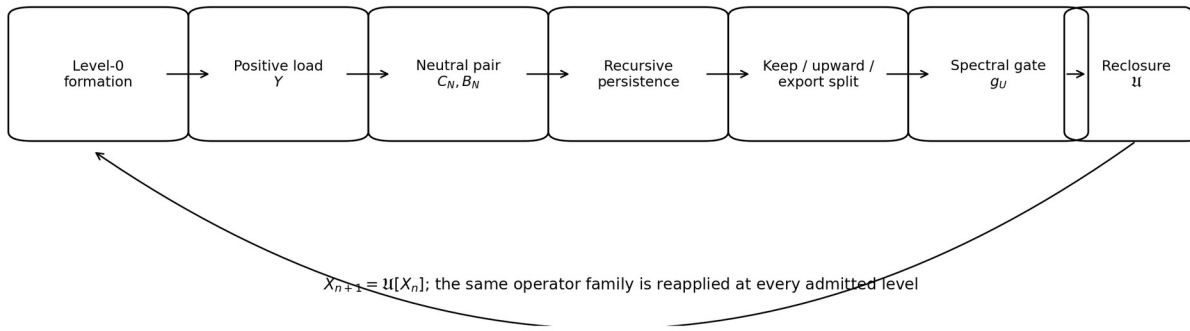


Figure 1. Universal reclosure architecture. The only iterated engine is the same sequence of formation, positive load, exact neutral resolution, persistence, retained/complement decomposition, spectral gating, and reclosure. The recurrence arrow denotes reapplication of the same operator family, not a new law at the next level.

5.2 Persistent witness projector

For every admitted recursive closure, persistence is defined spectrally before any target outcome is inspected. The witness projector is the Riesz projector around a prospectively declared isolated spectral block. In finite dimension this makes the persistent subspace unique relative to the chosen contour; under sufficiently small perturbations that do not close the contour gap, the projector varies continuously by the standard resolvent identity [7,8]. If the protected spectral block ceases to be isolated, the Strong Universal map does not move the contour to save the desired result; the iteration terminates into the non-generative object defined below.

$$\mathcal{W}_n = \{z: \sup_{k \geq 0} \|R_{\{C,n\}}^k z\| < \infty\} \quad (30)$$

$$P_{\{\mathcal{W}_n\}} = \text{RieszProj}_{\{\Sigma_p\}}(R_{\{C,n\}}) \quad (31)$$

The contour therefore belongs to the universal specification, not to a domain-specific fit. This converts spectral selection from an interpretive choice into a prospective structural condition and gives a precise failure mode for degeneracy or loss of robust persistence.

5.3 Canonical keep, upward, and export decomposition

The URE does not introduce an independent biological or cognitive information-gating operator. Instead it derives three operational roles from the same neutral pair and the same witness projector.

$$D_n^{\text{keep}} = P_{\{\mathcal{W}_n\}} C_N(Y_n) D_n \quad (32)$$

$$D_n^{\uparrow} = P_{\{\mathcal{W}_n\}} B_N(Y_n) D_n \quad (33)$$

$$D_n^{\text{out}} = (I - P_{\{\mathcal{W}_n\}}) B_N(Y_n) D_n \quad (34)$$

The three terms have different roles. D_n^{keep} is retained inside the current persistent closure; $D_n^{\uparrow}$ is complementary structure that is nevertheless carried by a persistent witness and is therefore available to participate in a higher relation; D_n^{out} is complementary structure that does not belong to the declared persistent witness block and is exported from the higher-generation packet. These are operator-defined roles, not new substances.

5.4 Canonical next-packet transport and positive-geometry preservation

The candidate packet is synthesized only from the retained persistent component and the persistent part of the complement. Let J_n denote the canonical isometric synthesis map associated with that ordered packet. The trinitarian capacity operators are transported by compression of their direct-sum lift through J_n . This transport rule is fixed once and is identical at every level; it therefore determines K_{n+1} from X_n rather than allowing an independently chosen next-stage formation matrix.

$$\tilde{D}_{n+1} = D_n^{\text{keep}} \oplus D_n^{\uparrow} \quad (35)$$

$$H_{n+1} = B_C[\tilde{D}_{n+1}]^\dagger W_N[\tilde{D}_{n+1}] B_C[\tilde{D}_{n+1}] \geq 0 \quad (36)$$

Positive master geometry is then preserved algebraically. The next closure Hessian is again a Gram-type positive operator $B_C^\dagger W_N B_C$ with W_N positive, and the master load is again $L_C^\dagger H_{n+1} L_C$. Consequently Y_{n+1} is positive semidefinite for every admitted packet without a new assumption about the sign of Y at the next level. This closes the positivity-preservation gap inside the declared finite-dimensional Strong Universal category.

5.5 Universal formation gate

Generation is decided by the same formation criterion at every iterate. The candidate K_{n+1} is already fixed by universal capacity transport; the gate therefore contains no stage-aware threshold. A simple isolated negative minimum mode admits a new formed branch. Loss of negativity, simplicity, or the required gap produces a terminal non-generative outcome rather than a retuned generator.

$$Y_{n+1} = L_C[\tilde{D}_{n+1}]^\dagger H_{n+1} L_C[\tilde{D}_{n+1}] \geq 0 \quad (37)$$

$$\begin{aligned} &(\mathcal{D}_{n+1}, c_{n+1}, \mathcal{K}_{n+1}) \\ &= J_n^\dagger [(\mathcal{D}_n, c_n, \mathcal{K}_n) \oplus (\mathcal{D}_n, c_n, \mathcal{K}_n)] J_n \\ &K_{n+1} = c_{n+1} + \mathcal{K}_{n+1} - \mathcal{D}_{n+1} \end{aligned} \quad (38)$$

Terminality is represented by a distinguished object $\perp$. This makes the universal map total: every admitted input has an output, but not every input generates a further nontrivial closure. The construction therefore distinguishes persistence, export, and higher generativity without forcing endless hierarchy.

5.6 Totalized Universal Reclosure Equation

The Universal Reclosure Equation now acts on the totalized state class. For an admissible state it constructs the persistent retained/complement packet, transports the same trinitarian capacities, rebuilds the positive master geometry, applies the same formation gate, and returns the next admissible state. If any mandatory admissibility condition fails, the image is $\perp$; and $\perp$ is absorbing. In this sense $\mathcal{U}$ is a genuine self-map of the totalized BFG state category rather than a partially defined family of sector transitions.

$$g_U(n) = \mathbb{1}_{\{\lambda_{\min}(K_{n+1}) < 0, \text{mult}(\lambda_{\min})=1, \Delta_{n+1} > 0\}} \quad (39)$$

$$X_{n+1} = \mathcal{U}[X_n] := g_U(n) C_N(Y_{n+1}) \tilde{D}_{n+1}, \text{ otherwise } \mathcal{U}[X_n] = \perp \quad (40)$$

$$\mathcal{U}: \tilde{X}_{\text{adm}} \rightarrow \tilde{X}_{\text{adm}}, \tilde{X}_{\text{adm}} := X_{\text{adm}} \cup \{\perp\}, \mathcal{U}(\perp) = \perp, X_n = \mathcal{U}^n[X_0] \quad (41)$$

Status of the URE
 Equations (36)–(38) are the new unification construction of this paper. They are built only from canonical BFG objects but the assertion that this reapplication is the correct universal cross-carrier generative rule is a theorem candidate / unification postulate, not an established fact of nature.

5.7 Endomorphism and no-hidden-retuning theorem

Universality is now stated without natural-stage predicates. Let X be an object of the totalized admissible category. A no-hidden-retuning failure occurs if the evaluation of $\mathcal{U}$ at X requires changing the neutral response family, the capacity-transport rule, the persistence contour rule, the order-consistency rule, the formation gate, or any numerical tolerance after the target domain is known.

$$I_U(X) := (\text{Spec } Y, \text{Spec } C_N, \text{Spec } B_N, \text{Spec } R_C, \dim \mathcal{W}, \varphi, \gamma_{\text{spec}}, F_{\text{nf}}, R_{\text{ret}}, H_{\text{nest}}) \quad (42)$$

$$X \sim_U X' \iff X \text{ and } X' \text{ are related by an admissible structure-preserving isomorphism with } I_U(X) = I_U(X') \quad (43)$$

Conditional theorem. Assume finite-dimensional admissible objects; bounded closure maps; positive W_N ; the exact neutral resolvent family; a prospectively isolated persistence block; functorial capacity transport through the canonical packet synthesis; and the common spectral formation rule. Then $\mathcal{U}$ maps the totalized admissible category into itself, preserves positive master geometry on every nonterminal iterate, and introduces no level-specific fundamental response law. The proof is by closure of each construction step: Gram operators preserve positivity, the resolvent pair is defined for positive load, the Riesz projector is defined on the isolated block, capacity transport produces the next K from the current object, and failure of any generativity condition maps to $\perp$ rather than to a repaired sector law.

6. Target-independent invariant classes and natural realization

The Strong Universal Form removes named stage predicates from the mathematics. After $\mathcal{U}$ generates an iterate, that iterate is represented only by a universal invariant signature containing spectral, persistence, order, factorization, return, and nesting data. Two states belong to the same mathematical class only when their invariant structures agree under an admissible structure-preserving isomorphism. The quotient is constructed before any comparison with physics, chemistry, biology, cognition, or phenomenology.

$$Q_U := \bar{X}_{\text{adm}} / \sim_U \quad (44)$$

$$\begin{aligned} \Pi_s: & \quad N_s \rightarrow Q_U, \\ \varepsilon_s(x) := & d_U(\Pi_s(T_s x), [\mathcal{U}(X_x)]_{\sim_U}) \end{aligned} \quad (45)$$

Natural names enter only through realization maps Π_s from independently described empirical systems into the universal quotient. A proposed atom, molecule, protocell, organism, conscious state, or qualitative content is therefore not generated because a predicate demanded that label. It is supported only if prospectively measured carrier data map into an already-existing invariant class and if observed transitions approximately commute with $\mathcal{U}$ under a fixed metric and tolerance. This removes the principal target-circularity risk while retaining a direct falsification route.

7. Physical closure from the same architecture

7.1 Local Physical Closure Principle

The Local Physical Closure Principle (LPCP) states that once an admissible closure enters a smooth, reversible, locally geometric regime, local propagation is the continuum limit of covariantly transported neighboring closure differences, using the same positive neutral stiffness that controls admissibility [1]. LPCP is a physical principle, not a Level-0 theorem.

$$\Delta_{\{\varepsilon, v\}} q(x) = U_C(x, x + \varepsilon v) q(x + \varepsilon v) - q(x) \quad (46)$$

$$\Delta_{\{\varepsilon, \nu\}} q / \varepsilon = \nabla_{\nu} \wedge C q + O(\varepsilon) \quad (47)$$

$$Z_C = H_C^{\{1/2\}} C_N(Y_C[q_*]) H_C^{\{1/2\}} > 0 \quad (48)$$

$$S_{LPC}^{\{(2)\}}[q] = \mathcal{K}_0 \int d^4x \sqrt{-g} [-\frac{1}{2} g^{\{\mu\nu\}} \langle \nabla_{\mu} \wedge C q, Z_C \nabla_{\nu} \wedge C q \rangle - V_C^{\{(2)\}}(q)] \quad (49)$$

$$V_C(q) = \frac{1}{2} \langle D_C(q), C_N(Y_C(q)) D_C(q) \rangle \quad (50)$$

The local response Hessian follows from derivatives of the same resolvent, rather than from an independent sector response.

$$\partial_A C = -C Y_{,A} C \quad (51)$$

$$\partial_A \partial_B C = C Y_{,A} C Y_{,B} C + C Y_{,B} C Y_{,A} C - C Y_{,AB} C \quad (52)$$

In the minimal single-metric, two-derivative branch, the principal symbol of every nondegenerate fundamental massless block is proportional to one metric null form. Under additional standard assumptions, the geometric gauge block is related to Fierz–Pauli massless spin-2 and the Einstein infrared branch [11–15]. These are conditional physical consequences, not additional BFG fundamental laws.

$$\sigma_2(\Gamma_a)(x, k) = Z_a g^{\{\mu\nu\}} k_{\mu} k_{\nu} P_a \quad (53)$$

$$g^{\{\mu\nu\}} k_{\mu} k_{\nu} = 0 \quad (54)$$

7.2 Physical realization map

A natural system is associated with the local physical branch only after the universal invariant class has been generated. The empirical realization map tests whether a smooth carrier, nondegenerate soldering, positive closure stiffness, reversible local transport, and stable characteristic structure correspond to that class under the Local Physical Closure Principle. These carrier conditions do not alter $\mathcal{U}$ and therefore cannot rescue a failed universal iterate.

Physical-stage

status

The existence of an admissible local geometric branch is conditional. The identification of the resulting metric and spectrum with observed spacetime and gravity remains empirical. The standard field-theory literature constrains the carrier interpretation [11–15]; it does not independently derive BFG.

8. Matter: nested nuclear, atomic, and molecular closure

Atomic and molecular systems are treated as realization tests of already-generated invariant classes, not as definitions of those classes. Established quantum mechanics, electrodynamics, nuclear physics, and chemistry supply the carrier observables [16]. BFG contributes the universal retained/complement, persistence, nesting, and export coordinates against which those observables are compared.

8.1 Atomic realization test

An atomic realization is supported only when an independently measured bound system maps to a universal class with an isolated persistent spectral block, noncollapsed distinguishability, bounded recursive transport, and admissible complement/export. The class exists mathematically without the word 'atom'; the label is attached only after the empirical map succeeds. No atomic Hamiltonian, threshold, or response law is inserted into $\mathcal{U}$.

$$\Pi_{\text{atom}}: N_{\text{atom}} \rightarrow Q_U \text{ is accepted only from independently measured carrier data} \quad (55)$$

This is intentionally a structural implication rather than a derivation of the hydrogen spectrum from a scalar toy load. The no-retuning claim is that the BFG closure law is unchanged; standard physics supplies the carrier Hamiltonian and measured spectral data.

8.2 Molecular and chemical reclosure

A molecular realization is tested by whether independently identified lower bound systems map jointly into a higher universal class whose retained response is nonfactorizable while the lower subclosures remain recoverable. The higher class is generated by $\mathcal{U}$ before chemical naming; chemistry supplies the carrier Hamiltonian and measured relational spectrum.

$$C_{\text{mol}} \neq C_{\text{atom},1} \oplus C_{\text{atom},2} \quad \text{when the higher retained response is nonfactorizable} \quad (56)$$

The distinction between constitutive nesting and temporal generation is important. The hierarchy expresses closure precedence. It does not imply that every molecule must literally replay Level-0 or atomic formation as a temporal developmental sequence.

9. From chemistry to abiogenesis

The molecular-to-biological transition is treated as a falsifiable generativity problem rather than an adjacent pair of labels [5]. External origin-of-life research shows that several carrier-level ingredients are experimentally tractable: activated pyrimidine ribonucleotides can arise under prebiotically plausible chemistry [17], common precursor chemistry can feed RNA, amino-acid, and lipid building blocks [18], template-directed polymer synthesis can occur in model protocells [19,20], protocell populations can exhibit competition [21], and recent autocatalytic systems can couple reaction dynamics to compartment growth and reproduction-like behavior [22]. These results constrain carrier plausibility; none validates BFG.

9.1 Prebiotic realization test

Reaction-network organization, compartment persistence, throughput, renewal, heredity, and repeated-cycle persistence are now treated strictly as empirical coordinates used to map prebiotic systems into the universal quotient. They are not terms in the generator and do not define a prebiotic mathematical class in advance. The only mathematical generator remains Equation (40).

$$\Pi_{\text{prebio}}: N_{\text{prebio}} \rightarrow Q_{\mathcal{U}} \quad (57)$$

$$\varepsilon_{\text{cycle}}(x) = d_{\mathcal{U}}(\Pi_{\text{prebio}}(T_{\text{cycle}} x), [\mathcal{U}(X_x)]_{\sim \mathcal{U}}) \quad (58)$$

For empirical work, the multicomponent observation vector $q_B, q_F, q_R, q_H, q_C, q_S$ remains useful [5], but its role is diagnostic. Those observables are measured independently and compared with invariant coordinates of $\mathcal{U}$ -generated classes. No component is permitted to change the universal capacity transport, neutral response, persistence rule, or gate.

$$\vec{q}_{\text{abi}} = (q_B, q_F, q_R, q_H, q_C, q_S) \quad (59)$$

A strong empirical test compares the full frozen BFG readout against capacity-matched hierarchy-only, compartment-only, heredity-only, and shuffled-coupling nulls. The repeated-cycle requirement remains: a one-off vesicle or isolated copying event is not enough to identify biological closure [5,23].

10. Cellular, multicellular, and organismic reclosure

10.1 Cellular closure

The previously described biological ordering is retained only as a natural-history and systems-biology comparison sequence [24]. In the Strong Universal mathematics there is no cellular, tissue, organ, or organism generator. There is only the orbit of $\mathcal{U}$ in the target-independent quotient; biological labels are empirical realization assignments made after the invariant classes exist.

A cellular realization therefore requires prospectively measured boundary persistence, throughput, informational renewal, and recursive self-maintenance across turnover to co-localize in one generated invariant class. Failure of that mapping is a failure of the proposed biological realization, not a reason to modify the universal operator.

10.2 Multicellular projection and morphogenesis

A prior organismic formulation treats local cellular closures as projecting into a form/field domain, being read out, differentiated, and reintegrated into a global closure [25]. Classical developmental biology provides independent mechanisms—reaction–diffusion, positional information, morphogen interpretation, gene-regulatory organization, mechanics, and bioelectric pattern control [26–30]. BFG does not replace these mechanisms. The unification claim is that they can be evaluated as carrier realizations of a common closure transition.

$$\Pi_{\text{org}}: N_{\text{org}} \rightarrow Q_U \quad (60)$$

A multicellular mass does not qualify merely because cell number increases. The organismic predicate requires persistent global integration, preserved local differentiation, admissible projection/readout, regulated export, and failure under the appropriate ablations. The distinction cell growth $\neq$ organism formation is retained [25].

10.3 Organismic closure

Organismic realization is treated as a many-subclosure empirical mapping problem. Local biological closures must remain individually recoverable while a higher nonfactorizable invariant class captures global persistence, differentiation, return, and regulated export. The organism label is assigned only after this higher class has been generated and matched; it never enters the formation rule.

11. Sensorimotor, cognitive, and memory closure

Sensorimotor, learning, memory, self-model, perspective, and identity concepts are likewise downstream realization hypotheses [24]. The universal mathematics instead tracks target-independent quantities such as externally driven distinguishability, cross-channel nonfactorization, recursive return, witness persistence, and self-referential reachability. Only after these coordinates are estimated independently may cognitive labels be attached.

Table 2. Target-independent cognitive invariant coordinates used only for downstream realization tests.

Invariant coordinate	Target-independent meaning
External distinguishability	Externally driven differences remain recoverable after neutral resolution and recursive transport.
Nonfactorization	Cross-channel closure cannot be reduced to a direct sum of independent retained blocks.
Return depth	Perturbations re-enter the same closure through a bounded sequence of return mappings.
Witness persistence	A prospectively selected invariant subspace remains power bounded across separated recursions.
Adaptive plasticity	Transverse degrees of freedom change while a declared witness remains recoverable.
Self-reference reachability	A closure representation can reach and be reached from a representation of its own persistent state.
World/self separation	Two distinguishable relational sectors remain jointly

	mediated without collapse.
Integration margin	Shared-neutral off-diagonal coupling remains nonzero under preserved differentiation.
Differentiation margin	The active retained operator keeps a positive minimum singular-value margin.

12. Thermodynamic maintenance as a readout, not a second consciousness law

A prior thermodynamic model introduced explicit entropy, information, and energy-balance equations [33]. In the present unification those equations are not used as a second fundamental definition of consciousness. Instead, thermodynamic quantities are read from a physical carrier that realizes the universal retained/complement partition.

For a normalized carrier state D , define retained and complementary quadratic fractions:

$$r_N(D) = \langle D, C_N(Y)D \rangle / \langle D, D \rangle \quad (61)$$

$$b_N(D) = \langle D, B_N(Y)D \rangle / \langle D, D \rangle = 1 - r_N(D) \quad (62)$$

A physical model may then map these fractions into bound energy, exported heat, chemical work, or information-processing costs. Landauer's result constrains the energetic cost of logically irreversible information operations [34], but it does not imply that consciousness is simply entropy reduction.

$$E_bind = \mathcal{R}_E^{\{bind\}}[r_N, carrier] \quad (63)$$

$$E_exp = \mathcal{R}_E^{\{out\}}[b_N, carrier] \quad (64)$$

The readout maps $\mathcal{R}_E$ are empirical constitutive maps, not new fundamental BFG dynamics: they do not feed back into $\mathcal{U}$ unless their measured carrier state changes Y in the already-declared upstream way. This preserves the distinction between a universal closure law and domain-specific thermodynamic units.

13. Conscious subjectivity as a shared-neutral URE class

The shared-neutral cognitive formulation already supplies a positive-load/resolvent family compatible with the universal architecture [10]. In the Strong Universal Form, however, 'conscious subjectivity' is not a mathematical stage predicate. It is an empirical realization hypothesis: a measured cognitive system must map into an invariant class exhibiting simultaneous shared-neutral integration, preserved differentiation, self/world structure, self-indexed witness continuity, and bounded recursion without changing $\mathcal{U}$.

$$Y_\varphi = M_\varphi \dagger M_\varphi \geq 0 \quad (65)$$

$$C_\varphi = (I + Y_\varphi)^{-1} \quad (66)$$

$$Q_\varphi = Y_\varphi (I + Y_\varphi)^{-1} = I - C_\varphi \quad (67)$$

Relative to a fixed channel decomposition with projectors P_i , exact channel factorization is equivalent for Y_φ and C_φ . Shared-neutral integration therefore requires nonzero off-diagonal response, while differentiation is protected by the smallest singular value.

$$Y_{\{ij\}}(\varphi) = P_i Y_\varphi P_j, \quad C_{\{ij\}}(\varphi) = P_i C_\varphi P_j \quad (68)$$

$$\Xi_\varphi = [\Sigma_{\{i \neq j\}} \|P_i C_\varphi P_j\|^2_{HS}] / \|C_\varphi\|^2_{HS} \quad (69)$$

$$\Delta_\varphi = \sigma_{\min}(C_\varphi|_{\{H_{act}\}}) = 1 / [1 + \lambda_{\max}(Y_\varphi|_{\{H_{act}\}})] \quad (70)$$

World/self relation, self-indexing, identity persistence, and bounded recursion therefore remain jointly testable coordinates, not definitional labels. The empirical identification of a matching invariant class with conscious subjectivity remains a hypothesis [10].

$$K_\varphi = R_\varphi C_\varphi \quad (71)$$

$$K_\varphi \dagger L K_\varphi \leq L \quad (72)$$

$$\sup_n \|K_\varphi^n x\|_L < \infty \quad (73)$$

Perturbational-complexity and near-criticality research provide external benchmarks for integration-plus-differentiation and bounded dynamical regimes [40,41]. BFG does not identify its variables directly with PCI or with a generic criticality index; such mappings must be preregistered.

14. Phenomenal content and qualitative identity

Qualitative content is handled in the same order: first the universal equivalence class, then the phenomenological realization claim [42]. The formal object $Q[X]$ is a closure-equivalence class determined by invariant structure. Whether one such class corresponds to experienced redness, pain, or another quale is not built into the quotient and must be established independently.

14.1 BFG closure equivalence

Let $\mathcal{G}_{\text{adm}}$ be the set of admissible transformations that preserve the declared witness spectrum, the neutral-response spectra, the local order state within tolerance, and the restricted recursive transport. Define:

$$X \sim_{\text{BFG}} X' \Leftrightarrow \exists G \in \mathcal{G}_{\text{adm}}: X' = GX \text{ and } \mathbb{I}_U(X') = \mathbb{I}_U(X) \quad (74)$$

$$\mathbb{I}_U(X) := \{\text{Spec } C_N(Y_X), \text{Spec } B_N(Y_X), \varphi_X, \mathcal{W}_X, R_C\{\mathcal{W}_X\}\} \quad (75)$$

$$Q[X] := [X]_{\sim_{\text{BFG}}} \quad (76)$$

The formal qualitative identity is therefore an equivalence class of the closure geometry, not an extra postulated substance. For two content closures:

$$Q_{\text{red}} \neq Q_{\text{blue}} \Leftrightarrow [C_{\text{red}}]_{\sim_{\text{BFG}}} \neq [C_{\text{blue}}]_{\sim_{\text{BFG}}} \quad (77)$$

This construction closes the formal operator gap only at the mathematical level. The empirical claim that a particular equivalence class is phenomenal redness still requires content-specific neural and behavioral tests. If closure invariants fail to distinguish red from blue after controlling local activation, or if the equivalence relation can only be defined using the target report itself, the BFG qualia claim is weakened or fails [42].

14.2 Ownership and selfhood

Ownership is likewise formulated as a relation between a content-equivalence class and a persistent identity witness after both have been generated within the universal quotient. The phrase 'my experience' is therefore an empirical interpretation of a bidirectional closure relation, not a primitive term in $\mathcal{U}$.

$$\text{SelfOwned}(Q_i) = 1 \Leftrightarrow Q_i \text{ is bidirectionally reachable from the persistent identity witness } \mathcal{W}_{\text{id}} \quad (78)$$

External work on bodily self-consciousness supports the broader empirical fact that ownership, self-location, perspective, and multisensory self-processing can dissociate [43]. It does not validate the BFG selfhood operator; it supplies independent phenomena against which the closure architecture can be tested.

15. The complete cosmic precedence map

The cosmic precedence map is retained only as an empirical comparison scaffold. It does not define the order in which $\mathcal{U}$ must produce named natural categories. The mathematical orbit contains unlabeled invariant classes; physical, chemical, biological, cognitive, and phenomenal names are assigned afterward if independent carrier data realize them. Structural precedence therefore remains conceptually useful without contaminating the universal generator with the desired destination classes.

Table 3. Natural-system realization scaffold mapped onto target-independent URE invariant coordinates.

Index	Natural realization hypothesis	Independent carrier signature to be mapped into Q_U
0	Level-0 admissibility	Difference/coherence/neutral capacities and an isolated negative formation mode.
1	Primitive formed structure	Observed carrier candidate maps to a class with a realized Level-0 branch and finite positive load.
2	Neutral-resolved structure	Observed response is consistent with the exact C_N/B_N partition.
3	Persistent structure	Measured dynamics support a prospectively isolated power-bounded witness block.
4	Local physical realization	Independent carrier data satisfy the physical realization conditions associated with a generated class.
5	Bound physical realization	Measured bound structure maps to a spectrally isolated retained class with admissible export.
6	Nuclear/atomic realization	Quantum carrier data map to a nested bound class with persistent spectral separation.
7	Molecular realization	Chemical data map multiple lower bound realizations into a higher nonfactorizable class.
8	Macromolecular realization / sequence	Measured relational ordering persists across reclosure-relevant perturbations.
9	Informational realization	Alternative sequence states remain discriminable and transportable under fixed carrier mapping.
10	Replication realization	A closure-relevant witness is

		reproduced into an additional same-level realization.
11	Error-correction realization	Perturbed carrier states return to the same target-independent invariant class.
12	Metabolic throughput realization	Persistent retention coexists with regulated measured export and renewal.
13	Boundary realization	Measured inside/outside partition remains stable while selective exchange persists.
14	Protocellular realization	Boundary, throughput, renewal, heredity, coupling, and repeated-cycle persistence co-map to one class.
15	Cellular realization	A protocellular carrier realization remains recursively self-maintaining across material turnover.
16	Projection realization	Measured persistent complementary structure becomes accessible to another local realization.
17	Readout realization	Projected structure measurably changes another closure-mediated carrier state.
18	Feedback realization	Projection and return form a bounded empirically observed recursive loop.
19	Network / field realization	Multiple local realizations map jointly into a nonfactorizable higher class.
20	Attractor realization	Perturbed carrier states return to one invariant basin under fixed measurement rules.
21	Differentiation realization	Distinct local basins persist while co-mapping to one higher class.
22	Tissue realization	Many cellular realizations support a persistent spatial-functional higher mapping.
23	Organ realization	Specialized tissue realizations map into a higher persistent functional

		class.
24	Organ-system realization	Multiple organ realizations jointly map into an integrated higher class.
25	Organism realization	Global empirical organization persists despite internal turnover and differentiation.
26	Sensory realization	External perturbations map to discriminable internal retained coordinates.
27	Integration realization	Multiple channels map to a shared nonfactorizable neutral-load class.
28	Valence realization	Alternatives exhibit reproducibly distinct persistence/export gradients.
29	Action realization	Measured export changes the exterior and returns as structured input.
30	Learning realization	Future measured transition structure changes after prior perturbation/feedback.
31	Memory realization	A witness remains recoverable across temporally separated cycles.
32	Self-model realization	Carrier data support a persistent internal representation of the system's own state.
33	Perspective realization	World-related and self-related coordinates remain distinct but jointly mediated.
34	Identity realization	A persistent self-index remains while transverse carrier degrees adapt.
35	Conscious-subjectivity realization hypothesis	Independent cognitive data map jointly to integration, differentiation, self/world structure, identity, and bounded recursion.
36	Phenomenal-unity realization hypothesis	Multiple content-related realizations occupy one globally connected differentiated class.
37	Qualitative-content realization hypothesis	Content identity is tested against a target-independent closure-equivalence class.

38	Self-owned experience realization hypothesis	Content-related class is bidirectionally linked to an independently estimated identity witness.
39	Narrative / intersubjective realization hypothesis	Temporally extended self-model and other-subject relations are tested only after class generation.
40	Knowledge / science / civilization realization hypothesis	Collective carrier systems are mapped post hoc into higher invariant classes; highly speculative empirical extension.

16. Executed numerical demonstrations

All simulations in this section were executed for the present whitepaper in Python 3.13.5 using NumPy 2.3.5 and Matplotlib 3.10.8. They are deterministic and dimensionless. No empirical dataset was fitted. The simulations are designed to test algebraic or stability consequences of the declared equations and to expose failure surfaces; they do not demonstrate that any physical, biological, or conscious system realizes BFG.

16.1 Exact neutral-partition surface

For two independent eigenloads $y_1, y_2 \geq 0$, define $c_i = (1 + y_i)^{-1}$ and $b_i = y_i / (1 + y_i)$. The exact partition $c_i + b_i = 1$ holds independently on each eigenmode. A symmetric diagnostic of simultaneous retained/complement availability is:

$$\Gamma_{rc}(y_1, y_2) = \frac{1}{2} [4c_1b_1 + 4c_2b_2], \quad 0 \leq \Gamma_{rc} \leq 1 \quad (79)$$

The numerical sweep used $y_1, y_2 \in [10^{-2}, 10^2]$. The maximum grid value was 0.999790 at $y_1 \approx 0.9715$, $y_2 \approx 0.9715$; the maximum absolute partition error was 2.220e-16, consistent with floating-point roundoff. The analytic maximum $\Gamma_{rc} = 1$ occurs at $y_1 = y_2 = 1$, where retained and complementary fractions are balanced mode by mode.

Exact Neutral Partition: Mean Eigenmode Reclosure Availability

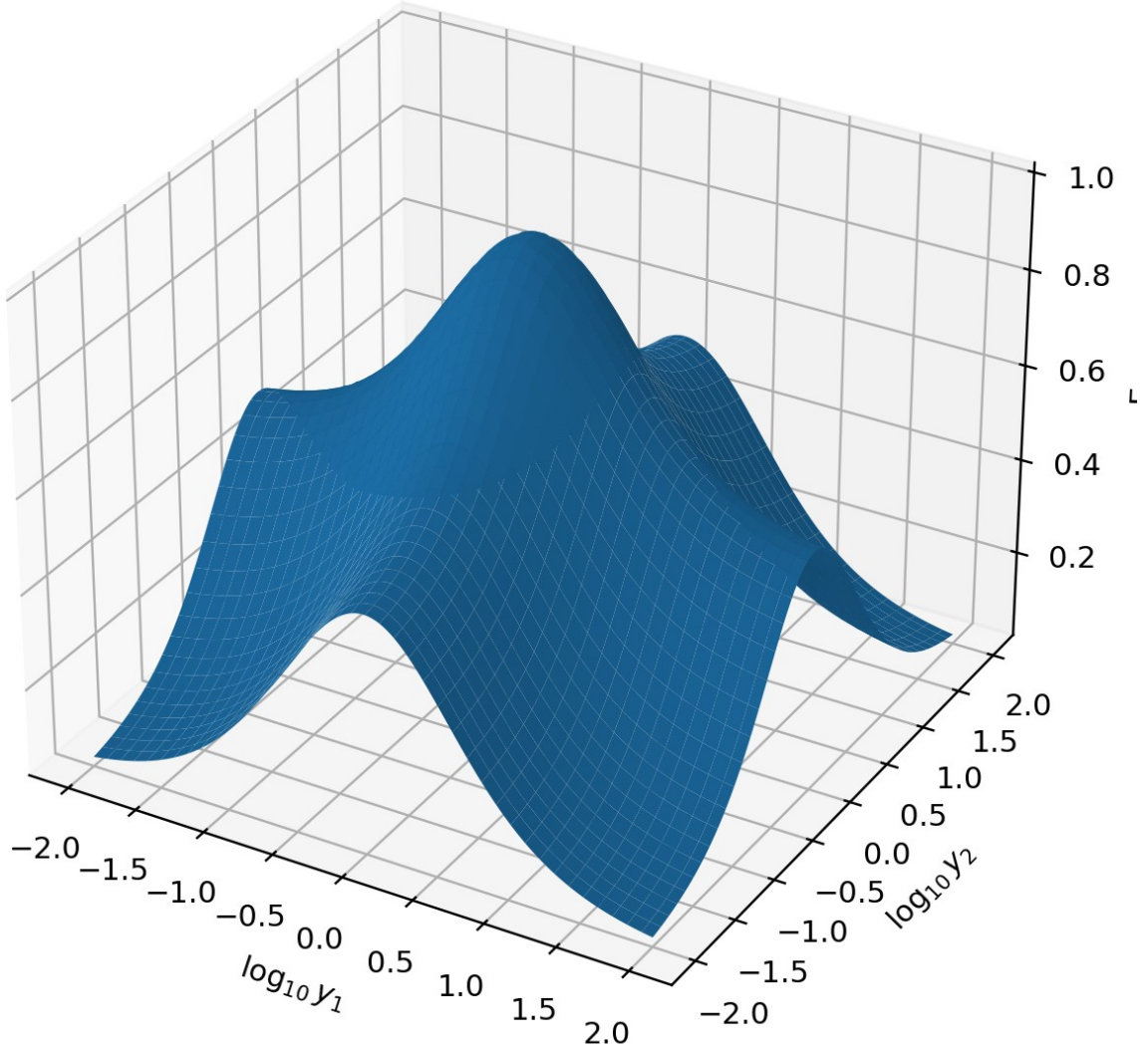


Figure 2. Executed 3D neutral-partition simulation. The surface is derived solely from the exact BFG resolvent pair. The ridge collapses to the analytic balanced maximum at $y_1=y_2=1$; no sector-specific response function is used.

16.2 Spectral generativity normal form

The spectral gate depends only on the sign of the next-level minimum formation eigenvalue. To visualize this sign change without assigning a physical sector, a two-mode symmetric normal form was used:

$$K_{nf}(\delta, g) = [[1+\delta, -g], [-g, 1-\delta]] \quad (80)$$

$$\lambda_{\min}(K_{nf}) = 1 - \sqrt{(\delta^2 + g^2)} \quad (81)$$

$$g_U = 1 \Leftrightarrow \delta^2 + g^2 > 1 \quad (82)$$

A dense sweep over $\delta \in [-1.6, 1.6]$ and $g \in [0, 1.6]$ reproduced the analytic circular sign boundary exactly to grid resolution. 69.5% of the displayed rectangular parameter window lies in the negative-eigenvalue region; this percentage has no physical meaning and depends only on the chosen plotting window. The scientific content is the visible gate surface $\lambda_{\min}=0$.

Spectral Generativity Gate: Two-Mode Normal Form

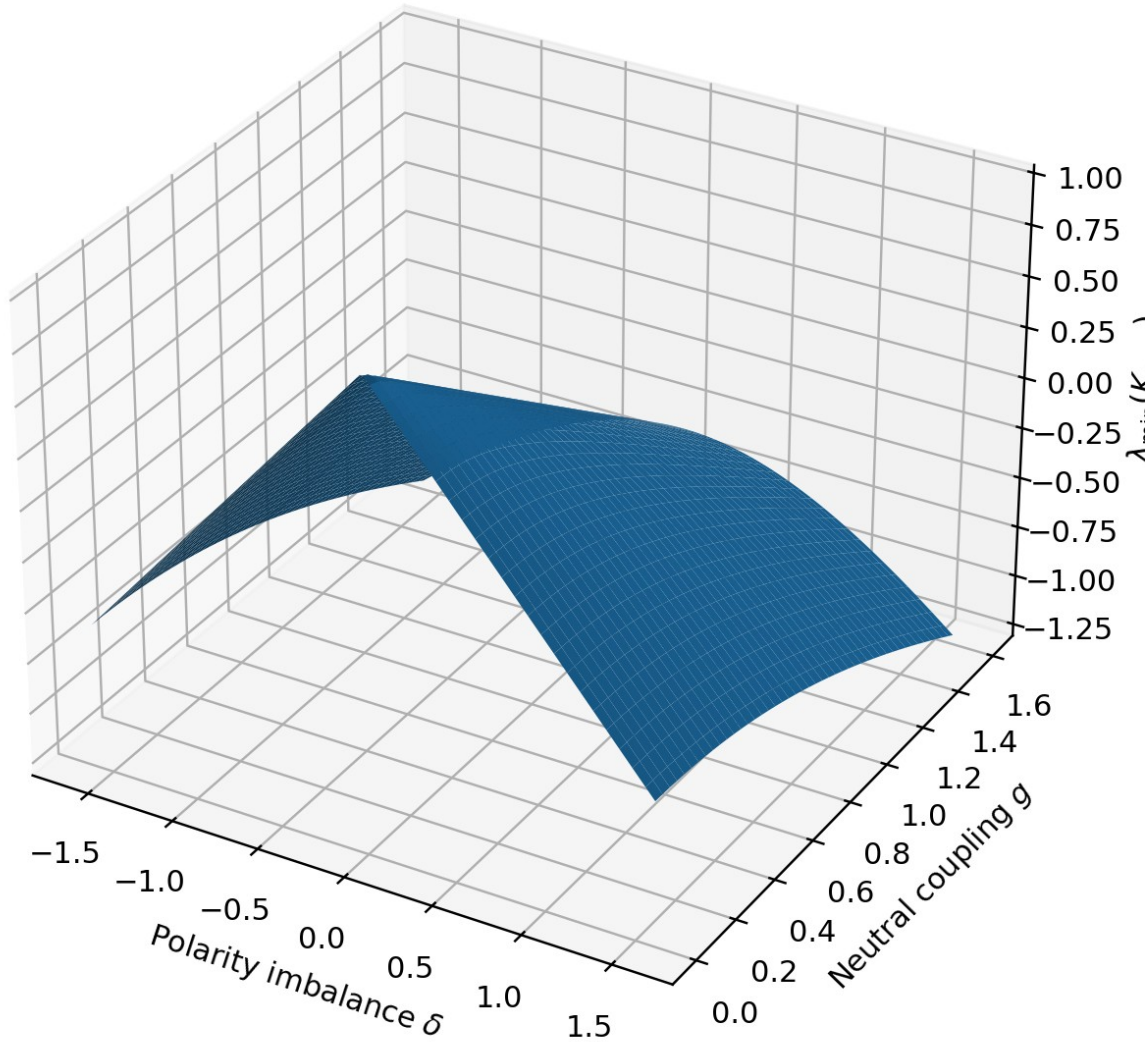


Figure 3. Executed 3D spectral-gate normal form. The surface illustrates the canonical rule $g_U=1$ when the candidate next-level formation operator acquires a negative minimum eigenvalue. The matrix is a pedagogical normal form, not a carrier law.

16.3 Order–feedback stability corridor

The declared order-indexed toy Jacobian uses $v(\varphi)=\exp[-4(\varphi-1)^2]$, $\alpha=0.35$, $\beta=0.25$, $d=0.45$, and $c=0.08$ [5]. The present work re-executed the full (φ, B) spectral-radius sweep.

$$v(\varphi)=\exp[-4(\varphi-1)^2] \quad (83)$$

$$J(\varphi, B) = [[B+0.35(1-v), 0.08v], [0.08v, 0.45+0.25(1-v)]] \quad (84)$$

At $B=0.82$ the executed values are $\rho(J) = 0.8366$ at $\varphi=1.0$ and 0.9897 at $\varphi=1.4$. At $B=0.90$ and $\varphi=1.4$, $\rho(J) = 1.0690$, crossing the linear stability boundary. Across the displayed $\varphi \in [0.45, 1.55]$, $B \in [0.55, 0.98]$ grid, 78.9% of sampled points had $\rho < 1$. That fraction is a plotting statistic, not a universal prediction.

Order-Indexed Stability Corridor

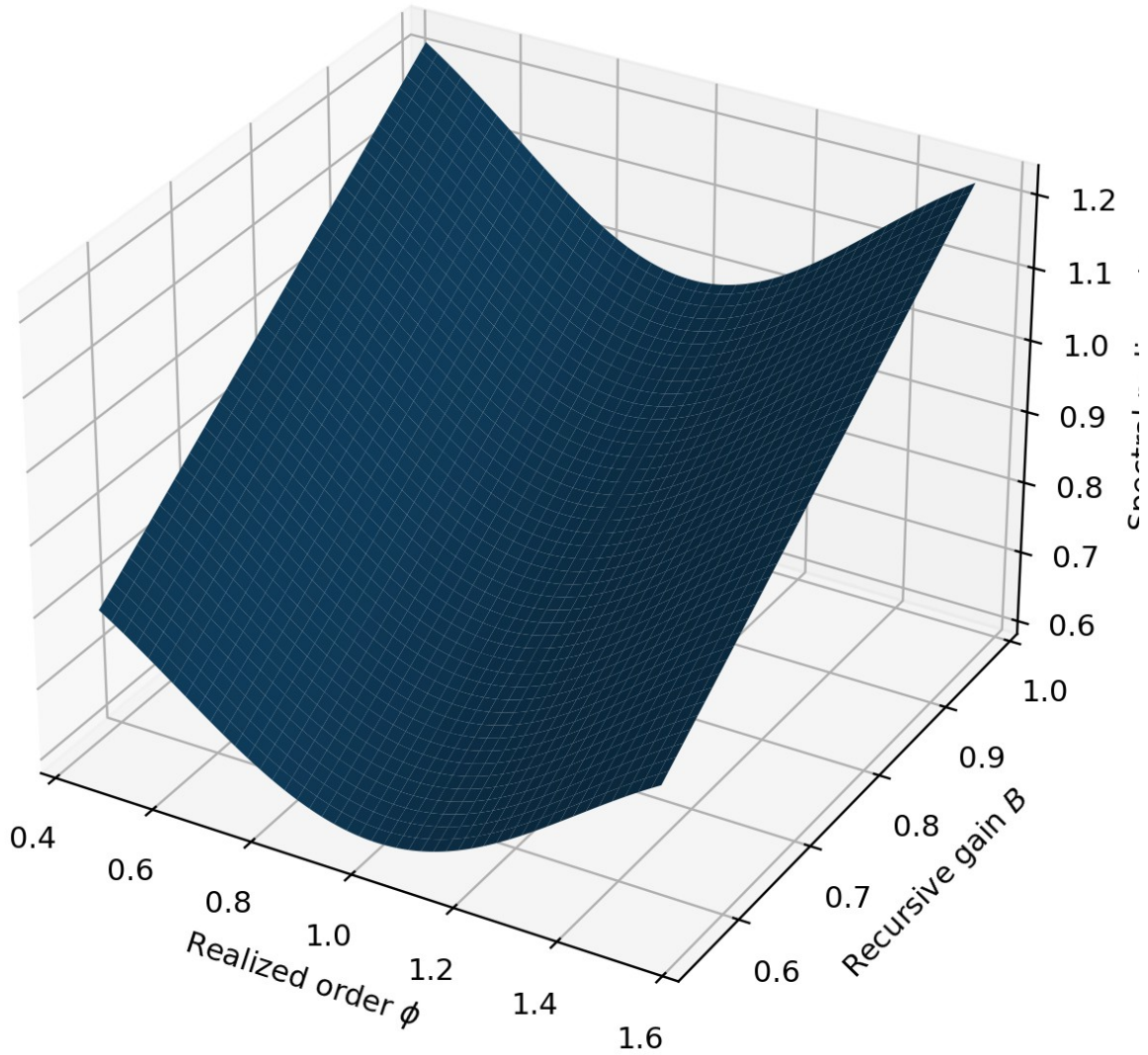


Figure 4. Executed 3D spectral-radius surface for the order-indexed interface toy model. The surface reproduces the finite joint stability corridor associated with the cited parameterization [5]. Linear spectral stability is necessary only for this fixed linear toy; nonnormal and time-varying systems require stronger power-bounded or Lyapunov analysis.

16.4 Shared-neutral cognitive integration

The five-channel cognitive simulation was re-executed from the previously defined matrices [10]. The mediation map uses $D = \text{diag}(0.55, 0.65, 0.75, 0.85, 0.95)$, a symmetric five-node ring with three weak chords normalized to spectral norm one, and $M_\varphi = D + g\varphi W$. With $\Phi_{\text{src}} = 1$ and $v(\varphi) = 1 + 0.45\varphi$, the positive order fixed point is:

$$\varphi^* = [-1 + \sqrt{1 + 4a\Phi_{\text{src}}}] / (2a) \quad (85)$$

The executed value is $\varphi^* = 0.7481$. At $g = 0, 0.4, 0.8, 1.2, 1.6$ the integration/differentiation pairs are respectively 0.0: (0.0000, 0.5256); 0.4: (0.0261, 0.4482); 0.8: (0.0819, 0.3442); 1.2: (0.1350, 0.2642); 1.6: (0.1697, 0.2061). These reproduce the previously defined trade-off [10]: coupling raises nonfactorizable integration while lowering the guaranteed differentiation margin.

Shared-Neutral Integration Surface

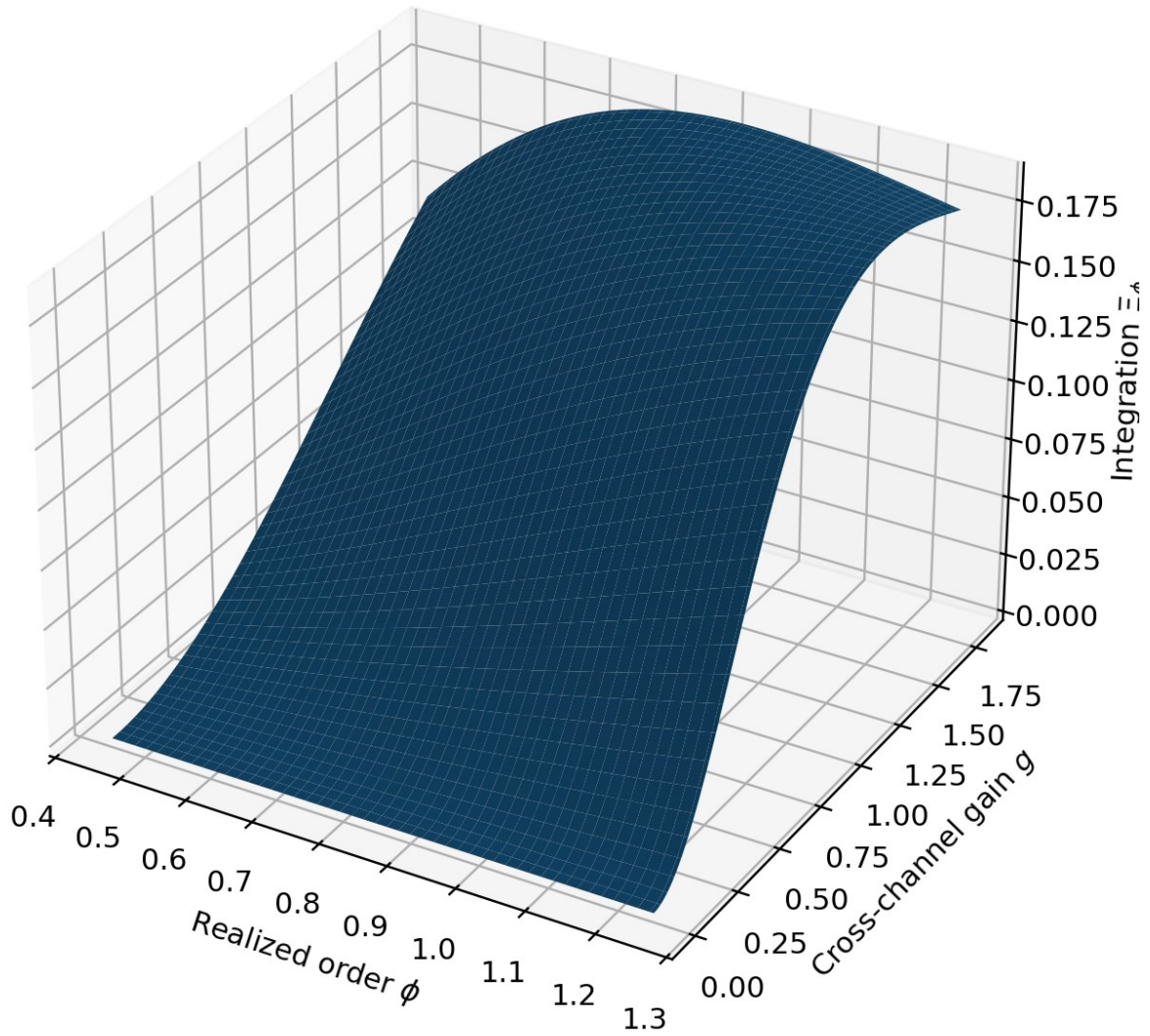


Figure 5. Executed 3D shared-neutral integration surface Ξ_ϕ . Integration vanishes in the $g=0$ ablation and grows as the common mediation map develops off-diagonal channel coupling. The plot is an internal mathematical demonstrator, not a consciousness measurement.

Preserved-Differentiation Surface

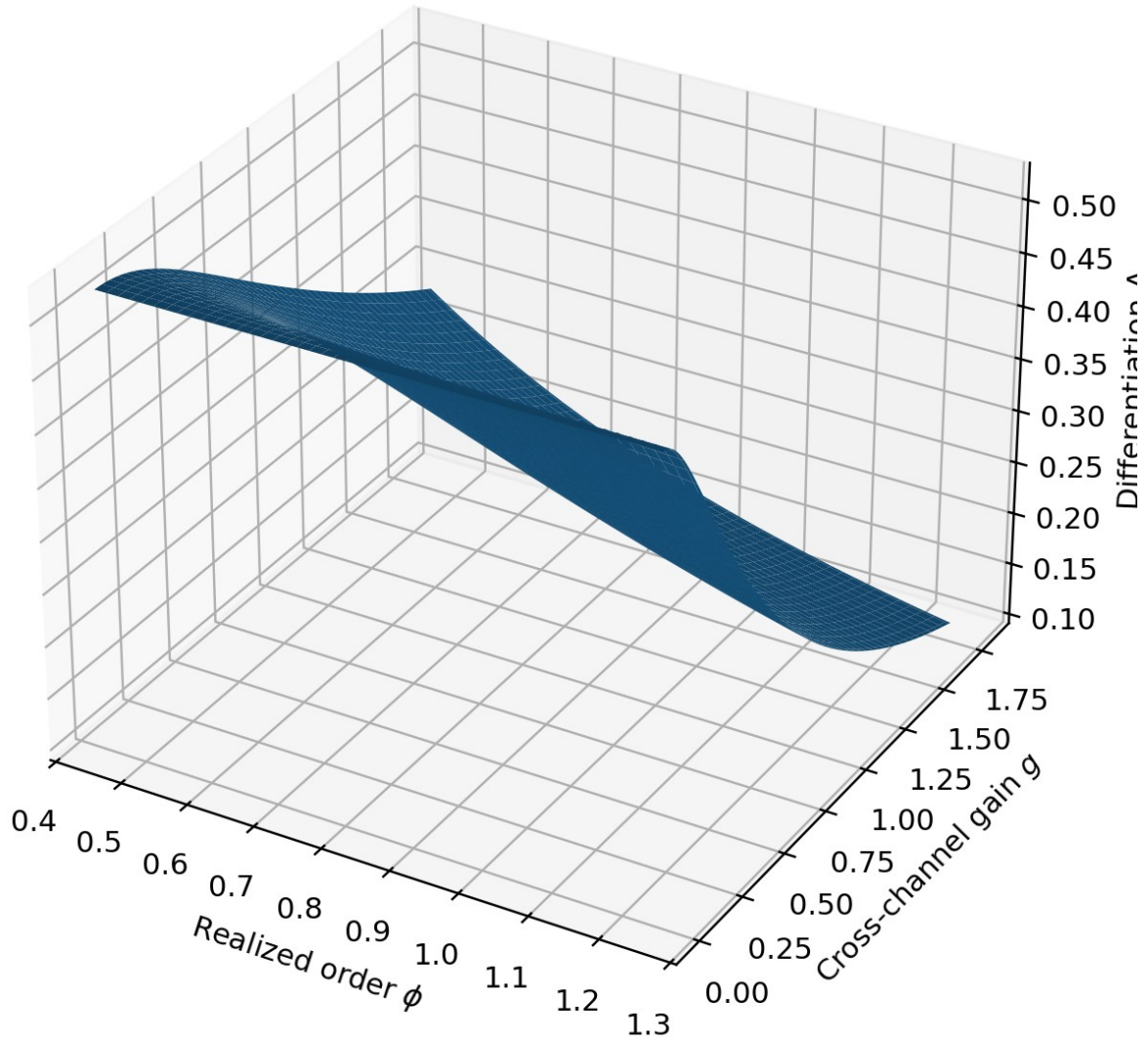


Figure 6. Executed 3D differentiation surface $\Delta_\phi = \sigma_{\min}(C_\phi)$ over the same parameter plane. Increasing shared load can erode the minimum distinguishability margin; this is why BFG does not identify maximal coupling with consciousness.

16.5 Recursive critical surface and derivative identity

At the benchmark $\phi^*=0.7481$ and $g=0.8$, the previous parameterization [10] uses $R_0=1 \oplus 0.96R(0.23) \oplus 0.86R(-0.37)$. The present execution gives:

$$K_\phi(r,g)=rR_0C_\phi(g) \tag{86}$$

$\rho(R_0C_\phi)=0.8711$, so $r_{\text{crit}}=1/\rho=1.1480$. Choosing $r=1.1422$ yields $\rho(K_\phi)=0.995000$. The analytic resolvent derivative $\partial_\phi C_\phi = -C_\phi Y'_\phi C_\phi$ was checked against a centered $h=10^{-6}$ finite difference with relative Frobenius error $4.297e-10$.

Recursive Stability Surface at Fixed Cross-Channel Gain $g=0.8$

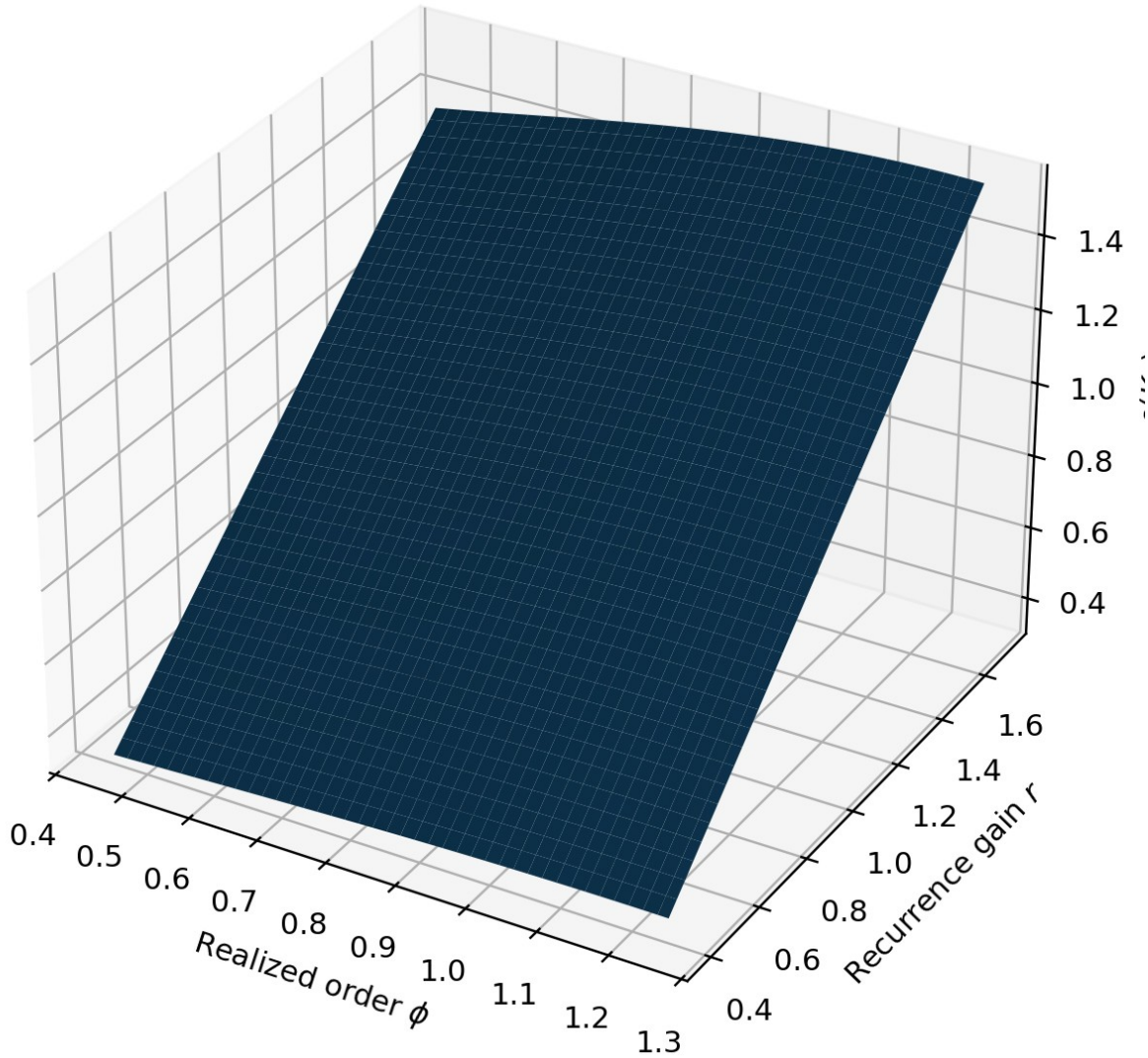


Figure 7. Executed 3D recursive stability surface for $K_\phi=rR_0C_\phi$ at fixed $g=0.8$. The critical level $\rho=1$ separates spectrally subcritical and supercritical regions for this fixed linear model.

16.6 Bounded near-critical trajectory

Using the near-critical gain above and the deterministic initial state $s_0=(0.90,-0.55,0.35,0.25,-0.20)$, 160 iterations remained bounded. The initial Euclidean norm was 1.1565, the maximum norm over the run was 1.3535, and the final norm was 0.6575. The trajectory is shown in the first three principal coordinates of the 161 simulated states.

Bounded Near-Critical Recursive Trajectory (PCA Projection)

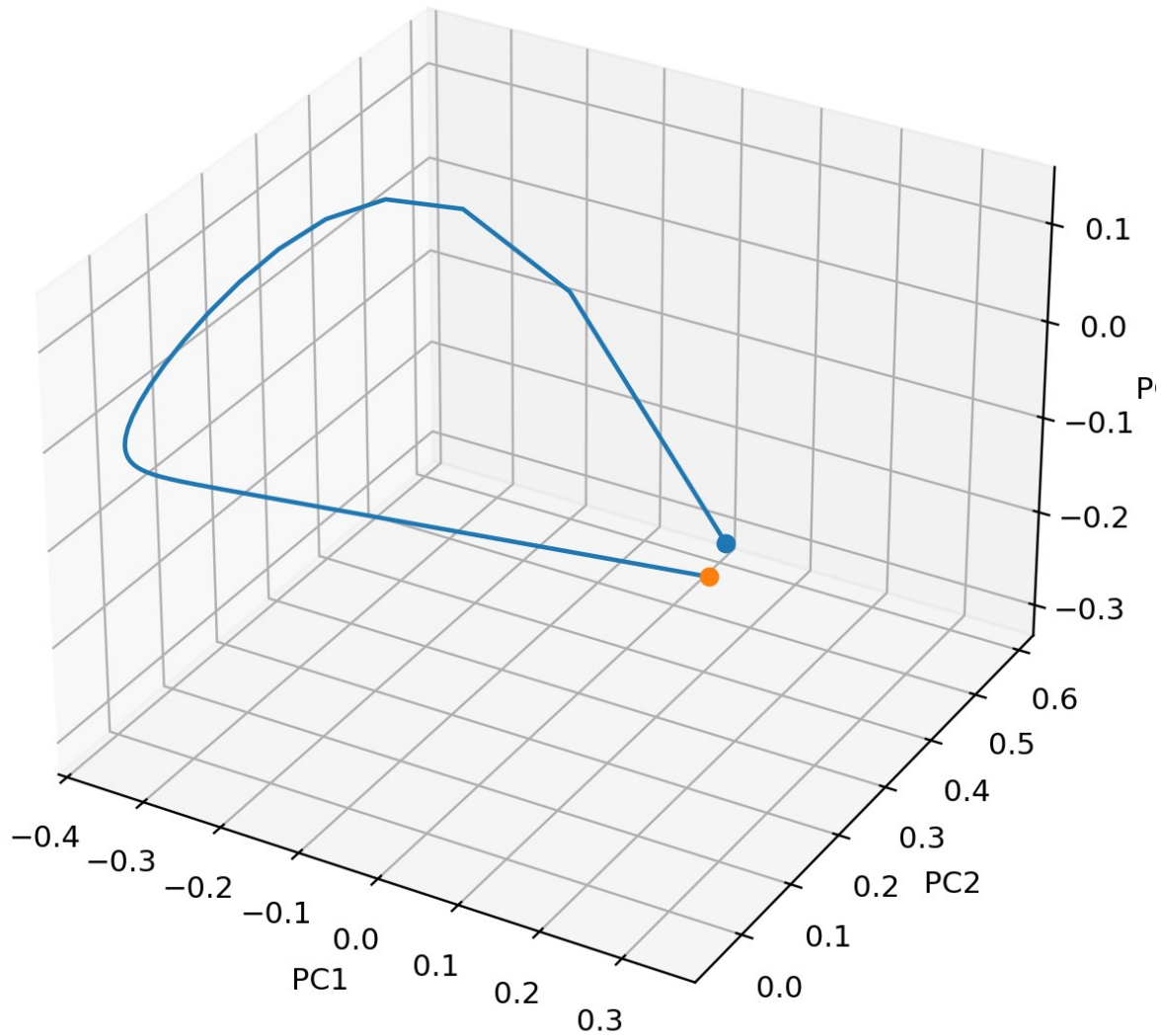


Figure 8. Executed bounded near-critical trajectory projected into three principal coordinates. The run demonstrates power-bounded recurrent behavior for the chosen fixed linear model. It does not demonstrate phenomenal consciousness.

17. Interpretation of the simulations

The simulations support internal consequences of the Strong Universal architecture only. They verify the exact neutral partition numerically, display a target-independent spectral formation boundary, reproduce the declared order-feedback stability corridor, and show that one shared-neutral positive-load family can simultaneously carry integration, differentiation, and bounded recursive dynamics. None of the simulations uses an atom-, cell-, organism-, or consciousness-specific generator.

The simulations do not prove natural universality. Their role is to establish calculational coherence and expose failure surfaces for the universal operator itself. The stronger empirical test is whether independent natural-system measurements map into the same invariant quotient and whether observed transitions commute with $\mathcal{U}$ within prospectively fixed error bounds.

18. Falsification architecture

A universal mathematical theory becomes vacuous if a failed realization can always be repaired by modifying the generator or by redefining the target class after inspection. The Strong Universal Form therefore separates generator failures, quotient failures, realization failures, and empirical carrier failures. Only the first two threaten mathematical universality; the latter two test whether nature realizes the universal category.

Table 4. Primary falsification surfaces of the URE program.

Failure surface	Decision condition
Formation failure	$\lambda_0 \geq 0$ or the Level-0 negative mode is not isolated.
Positive-geometry failure	A declared Gram-type construction or positivity domain fails, or $Y_{\{n+1\}}$ cannot be reconstructed as positive without changing the universal rule.
Neutral-resolution failure	Empirical response requires an independent pole or nonlinear law not derivable from the universal C_N/B_N family.
Persistence failure	The declared witness block is not power bounded or moves only under target-aware spectral selection.
Generativity failure	A higher nonterminal iterate requires a target-specific K , gate, threshold, or transport map not fixed by the universal construction.
Carrier-transfer failure	The empirical realization map must alter the universal generator or invariant quotient after entering a new domain.
Hierarchy-only null	A capacity-matched generic hierarchy predicts the same held-out transitions without BFG mediation/return structure.
Atomic failure	The proposed atomic readout adds no predictive or organizing content beyond relabeling established carrier theory.
Abiogenesis failure	Repeated-cycle closure does not depend on the predicted coupling, export, or witness relations under ablation.
Organism failure	Projection/readout/mediation variables add no prospective value beyond matched developmental nulls.
Consciousness failure	Matched recurrent or connectivity models reproduce held-out perturbational predictions without shared-neutral constraints.
Qualia failure	No non-circular content-specific closure invariant is found, or local carrier activity fully explains content and ownership.
Global no-retuning failure	Any target domain can be fit only by modifying the universal generator, invariant coordinates, quotient metric, or tolerance after seeing the outcome.

19. Preregistration and no-retuning protocol

Every empirical realization test must freeze, before target outcomes are inspected, the carrier-to-BFG role extraction, universal capacity-transport rule, neutral response family, persistence contour rule, order solver, invariant metric, realization tolerance, and all ablation/null comparisons. Natural labels are withheld during construction of the invariant classes and introduced only after the map is frozen.

- Freeze operator family and numerical tolerances before outcome access.
- Separate structural level n from physical/developmental time t .
- Report retention, complement, export, spectral margin, and witness continuity jointly.
- Include hierarchy-only, order-independent, return-ablated, shuffled-coupling, and capacity-matched recurrent controls where applicable.
- Preserve negative outcomes and localize the failed bridge instead of rewriting the stage definition.
- For consciousness and qualia, preregister report-independent or report-minimized carrier estimators and construct the invariant class without using the phenomenal label as a feature.
- For abiogenesis, require repeated-cycle persistence rather than a one-off successful compartment.
- For cross-domain claims, use the same universal quotient coordinates and generator; only the empirical carrier map Π_s may differ, and its form must be declared before target outcomes are inspected.

20. Relationship to established science

The Strong Universal Form is a mathematical closure architecture and a natural-realization research program, not a replacement for mature carrier theories. Matrix analysis and perturbation theory constrain the positive and spectral claims [7,8]; fixed-point theory constrains order consistency [9]; established physics constrains physical realization; chemistry and origin-of-life research constrain prebiotic realization; developmental biology constrains organismic realization; and neuroscience constrains cognitive and phenomenal realization. Those theories supply empirical carrier structure, while $\mathcal{U}$ supplies the proposed domain-independent closure grammar.

The burden is therefore sharper than simple explanatory compatibility. BFG must demonstrate that one prospectively frozen invariant quotient and one self-mapping operator compress, predict, or discriminate held-out natural structure across domains better than matched alternatives, without target-aware retuning. Failure of that cross-domain realization does not invalidate the internal endomorphism theorem, but it does invalidate the stronger claim that nature broadly realizes the BFG universal category.

21. Boundary of mathematical universality and natural realization

The Strong Universal Form closes the principal internal gaps identified for the earlier construction: the next formation operator is no longer independently selected; positive master geometry is rebuilt by a positive Gram construction; persistence selection is prospectively spectral; the map is totalized with an absorbing non-generative object; and named natural stages are removed from mathematical classification. What remains open is not the internal definition of mathematical universality over the declared category, but the breadth and uniqueness of its realization in nature.

Table 5. Open proof obligations after formal unification.

Boundary / realization obligation	Strong Universal status
Foundational source uniqueness	The Strong Universal category is mathematically defined, but Level-0 capacity operators are not claimed to be uniquely forced by a deeper empirically established theory.
Natural representation maps	Π_s is intentionally external to the generator. Existence, uniqueness, and measurement robustness must be established separately for each empirical carrier family.
Natural-class existence	The quotient can be generated without target labels; it does not force every named natural regime to occur for

	generic initial data.
Atomic realization	Actual spectra and constants remain supplied by established quantum carrier theory; the test is whether those data map into the universal quotient without retuning.
Abiogenesis realization	The historical terrestrial pathway is not derived; repeated-cycle prebiotic data test whether relevant systems realize the predicted invariant relations.
Morphogenetic realization	Gene regulation, mechanics, morphogens, bioelectricity, and extracellular matrix dynamics remain carrier mechanisms whose relation to universal classes is empirical.
Consciousness realization	A shared-neutral invariant class does not by itself establish phenomenality; independent neural and behavioral realization tests are required.
Qualia realization	Target-independent equivalence removes a free formal content label, but mapping an invariant class to reported qualitative content remains empirical.
Thermodynamic realization	Energy and entropy units require carrier-specific constitutive measurement maps that must not modify $\mathcal{U}$.
Higher collective realization	Narrative, intersubjective, scientific, and civilizational mappings remain substantially more speculative and are not needed for the mathematical universality theorem.

22. Universal realization and validation program

The validation program now has a strict order. First, publish the universal state schema, capacity-transport rule, persistence contour rule, invariant signature, quotient metric, and all numerical tolerances as frozen mathematical objects. Second, construct invariant classes without natural labels. Third, define independent carrier maps Π_s for physical, chemical, biological, and cognitive datasets. Fourth, test whether measured transitions approximately commute with $\mathcal{U}$. Fifth, compare predictive compression and perturbational performance against capacity-matched null models. Only after these steps may a generated class be interpreted as an atom, cell, organism, conscious state, or phenomenal content.

The success criterion is consequently domain transfer without generator retuning. Mathematical universality is the endomorphism property of $\mathcal{U}$ on the totalized admissible category. Natural universality requires multiple independently measured domains to realize that same quotient and transition law under fixed rules. These two claims are kept separate so that success in one cannot be used to conceal failure in the other.

23. Strong Universal Self-Closure Theorem

Conditional	Theorem	—	Strong	Universal	Self-Closure
Assume: (i) a simple isolated negative Level-0 formation mode; (ii) bounded finite-dimensional admissible objects; (iii) exact neutral elimination with $C_N + B_N = I$; (iv) positive Gram-type closure construction $H_C = B_C^\dagger W_{NB_C}$					

with $W_N \geq 0$ and $Y_C = L_C^\dagger H_{CL_C}$; (v) a prospectively isolated power-bounded persistence block with a fixed Riesz contour rule; (vi) canonical packet synthesis and functorial transport of the trinitarian capacity operators; (vii) the common order-consistency rule; and (viii) one common spectral formation gate. Then the totalized Universal Reclosure operator $\mathcal{U}: \tilde{X}_{\text{adm}} \rightarrow \tilde{X}_{\text{adm}}$ is a self-map, preserves positive master geometry on every nonterminal iterate, and requires no stage-specific fundamental response law. Natural labels are introduced only through downstream realization maps Π_s into the target-independent quotient Q_U . The theorem establishes mathematical universality over the declared BFG category; it does not establish that every natural system realizes that category.

Proof sketch. Let X belong to the totalized admissible state category. If $X = \perp$, the absorbing definition gives $\mathcal{U}(X) = \perp$. Otherwise the exact neutral pair exists because $Y \geq 0$, the prospectively isolated persistence block defines P_W , and the keep/up/export decomposition is therefore unique relative to the frozen contour. The candidate packet is synthesized from keep and persistent complement only. The trinitarian capacity operators are transported into that packet by the same fixed compression rule, so K_{n+1} is determined by X_n rather than selected for a target stage. The next closure Hessian is $B_C^\dagger W_{NB_C} \geq 0$ and therefore $Y_{n+1} = L_C^\dagger H_{CL_C} \geq 0$. If the common formation criterion passes, $C_N(Y_{n+1})$ is well defined and the result is another admissible object; if the criterion or a mandatory gap condition fails, the image is $\perp$. Hence $\mathcal{U}$ is a self-map of the totalized admissible category. Because natural labels occur only in the downstream realization maps Π_s and not in $\mathcal{U}$, no target-specific stage law or stage predicate participates in generation. This proves mathematical universality over the declared BFG state category, conditional on the stated structural axioms and model-completion choices. It does not prove that every natural system realizes that category.

24. Conclusion

The Strong Universal Form turns the cross-domain BFG program into a mathematically self-closed architecture. Level-0 supplies the initial admissible trinitarian structure. Exact neutral elimination supplies the retained/complement pair. Recursive conjugate closure supplies persistence. Functorial capacity transport determines the next formation operator. Gram-type reconstruction preserves positive master geometry. The universal spectral gate decides whether higher formation occurs, and the terminal object makes the recursion total without forcing artificial continuation.

The decisive change is that atom, molecule, protocell, cell, organism, cognition, conscious subjectivity, qualitative content, and self-owned experience are no longer mathematical stage predicates. The operator first produces domain-independent invariant classes in a universal quotient. Only afterward may independent natural-system data be mapped into those classes. This removes the desired target from the generator and makes cross-domain realization genuinely testable.

The resulting universality claim is exact but bounded: $\mathcal{U}$ is universal over the declared admissible BFG category under one fixed operator family, one positive-load construction, one persistence rule, one capacity-transport rule, one order logic, and one failure accounting. Whether the physical universe, living systems, and conscious systems are all realizations of that mathematical category is a separate empirical hypothesis. A universal mathematics and a universal theory of nature are therefore connected by a falsifiable realization map rather than conflated by definition.

Appendix A. Core formula register

Table 6. Compact formula register.

Role	Formula / statement
Level-0 source	$\mathcal{T}_0 = (\mathcal{D}_0, c_0, \mathbf{N}_0); K_0 = c_0 + \mathbf{N}_0 - \mathcal{D}_0$
Formation	$\delta^*_{\pm} = \pm \sqrt{(-\lambda_0) c_0}$ for simple $\lambda_0 < 0$
Positive load	$Y = A^\dagger A \geq 0; Y_C = L_C^\dagger H_{CL_C}$
Neutral retention	$C_N(Y) = (I + Y)^{-1}$
Complement	$B_N(Y) = Y(I + Y)^{-1} = I - C_N(Y)$
Exact partition	$C_N + B_N = I$; quadratic retained+complement budget exact
Persistence	$R_{Ce_T} = e_T$; stable complement strictly contractive in a Lyapunov norm
Order fixed point	$\varphi_{nv_n}(\varphi_n) = \Phi_{src,n}$
Witness subspace	$\mathcal{W}_n = \{z: \sup_k R_{C,n^k z} < \infty\}$
Keep / upward / export	$P_{WC_{ND}}; P_{WB_{ND}}; (I - P_W)B_{ND}$
Candidate packet	$\tilde{D}_{\{n+1\}} = D_{keep} \oplus D_{up}$; canonical synthesis J_n transports the trinitarian capacities.
Spectral gate	$g_U = 1$ only for a simple isolated negative minimum eigenvalue of the universally transported $K_{\{n+1\}}$.
URE	$X_{\{n+1\}} = g_U C_N(Y_{\{n+1\}}) \tilde{D}_{\{n+1\}}$; failure maps to the absorbing object $\perp$.
Iteration	$\mathcal{U}: \tilde{X}_{adm} \rightarrow \tilde{X}_{adm}, \mathcal{U}(\perp) = \perp, X_n = \mathcal{U}^n[X_0]$.
Cognitive load	$Y_\varphi = M_\varphi^\dagger M_\varphi$
Cognitive integration	$\Xi_\varphi = \sum_{\{i \neq j\}} P_{iC_\varphi P_{jI}} ^2_{HS} / C_\varphi ^2_{HS}$
Differentiation	$\Delta_\varphi = \sigma_{\min}(C_\varphi H_{act})$
Universal invariant quotient	$Q_U = \tilde{X}_{adm} / \sim_U$; natural labels enter only through external realization maps Π_s .

Appendix B. Simulation reproducibility register

Table 7. Exact numerical specification for executed figures.

Item	Specification
Software	Python 3.13.5; NumPy 2.3.5; Matplotlib 3.10.8.
Neutral surface	y_1, y_2 logarithmically sampled from 10^{-2} to 10^2 ; 160×160 grid; $\Gamma_{rc} = \frac{1}{2}(4c_1 b_1 + 4c_2 b_2)$.
Order-feedback surface	$\varphi \in [0.45, 1.55]$; $B \in [0.55, 0.98]$; $\alpha = 0.35$, $\beta = 0.25$, $d = 0.45$, $c = 0.08$; 180×180 grid.
Spectral gate normal form	$\delta \in [-1.6, 1.6]$, $g \in [0, 1.6]$; $\lambda_{\min} = 1 - \sqrt{(\delta^2 + g^2)}$; 180×180 grid.

Cognitive W	Unit-weight ring edges (1,2),(2,3),(3,4),(4,5),(5,1); chords W13=0.35, W24=0.25, W35=0.30; normalized to $lWl_2=1$.
Cognitive D	diag(0.55,0.65,0.75,0.85,0.95).
Order fixed point	$\Phi_{\text{src}}=1$; $a=0.45$; $\varphi^*=0.7481333923$.
Integration surface	$\varphi \in [0.45, 1.25]$, $g \in [0, 1.8]$; 140×140 grid.
Recursive primitive	$R_0 = 1 \oplus 0.96R(0.23) \oplus 0.86R(-0.37)$.
Near-critical gain	$r = 0.995/p(R_0C_\varphi) = 1.1422297734$ at $\varphi^*=0.7481333923$, $g=0.8$.
Derivative check	Centered finite difference $h=10^{-6}$; relative Frobenius error 4.2967×10^{-10} .
Trajectory	Initial $s_0 = (0.90, -0.55, 0.35, 0.25, -0.20)$; 160 iterations; PCA obtained by SVD of centered 161-state matrix.

Appendix C. Source-credit register

Table 8. Source-line provenance used by the Unification Paper.

Source line	Use in present paper
Marcel Theodor Wendé & Thomas W. Loker [3]	Co-authored Level-0 admissibility and pre-polar formation source line.
Marcel Theodor Wendé & Luke Kenneth Casson Leighton [4]	Recursive conjugate-closure source line $R_C = MRM^\dagger$, $M = \sqrt{N}$ and related stability geometry.
Canonical BFG source [1]	Universal positive closure architecture, exact neutral elimination, spectral gate, and LPCP.
Prior BFG sources [5,6]	Local order, fixed-point mediation, generativity/return architecture, and cross-scale compatibility.
Prior carrier-specific BFG sources [10,16,24,25,33,42]	Carrier-level structural constraints for atomic, biological, organismic, cognitive, phenomenal, and thermodynamic readouts.
This paper	Strong Universal Reclosure Equation, totalized admissible category, functorial capacity transport, target-independent invariant quotient, natural-realization tests, and executed synthesis simulations.

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Declarations

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Author contribution. Conceptualization, BFG unification architecture, theoretical formulation, interpretation, and manuscript responsibility: Marcel Theodor Wende. The Level-0 and recursive conjugate-closure source lines used in the manuscript are credited through the numbered bibliography.

Data availability. No external empirical dataset was analyzed. Numerical results are synthetic deterministic calculations from equations and parameters fully stated in the manuscript.

Code availability. The simulation equations, parameter ranges, software versions, and deterministic initial conditions required to reproduce the displayed figures are provided in Appendix B. A public repository release should include the exact script and environment lockfile before journal submission.

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